We'll guide you through these steps with the help of a CloudFormation template. You can find the CloudFormation template for this example on GitHub and on S3. Download a snapshot of the repository at http://github.com/AWSinAction/code3/archive/master.zip. The file we're talking about is located at chapter13/recovery.yaml. On S3, the same file is located at https://s3.amazonaws.com/awsinaction-code3/chapter13/recovery.yaml.

The following command creates a CloudFormation template that launches an EC2 instance with a CloudWatch alarm that triggers a recovery if the virtual machine fails. Replace `$Password` with a password consisting of 8–40 characters. The template automatically installs a Jenkins server while starting the virtual machine:

```
$ aws cloudformation create-stack --stack-name jenkins-recovery \
➥ --template-url https:/ /s3.amazonaws.com/\
➥ awsinaction-code3/chapter13/recovery.yaml \
➥ --parameters "ParameterKey=JenkinsAdminPassword,
➥ ParameterValue=$Password" \
➥ --capabilities CAPABILITY_IAM
```

While you are waiting for CloudFormation to launch the EC2 instance, we will have a look into the CloudFormation template. The CloudFormation template contains the definition of a private network. The most important parts of the template follow:

- A virtual machine with user data containing a bash script, which installs a Jenkins server during bootstrapping
- A public IP address assigned to the EC2 instance, so you can access the new instance after a recovery using the same public IP address as before
- A CloudWatch alarm based on the system status metric published by the EC2 service

The next listing shows how to create an EC2 instance that runs a script to install Jenkins during bootstrapping. The Elastic IP address ensures a static public IP address for the virtual machine.

Listing 13.1 Starting an EC2 instance using a Jenkins CI server with recovery alarm